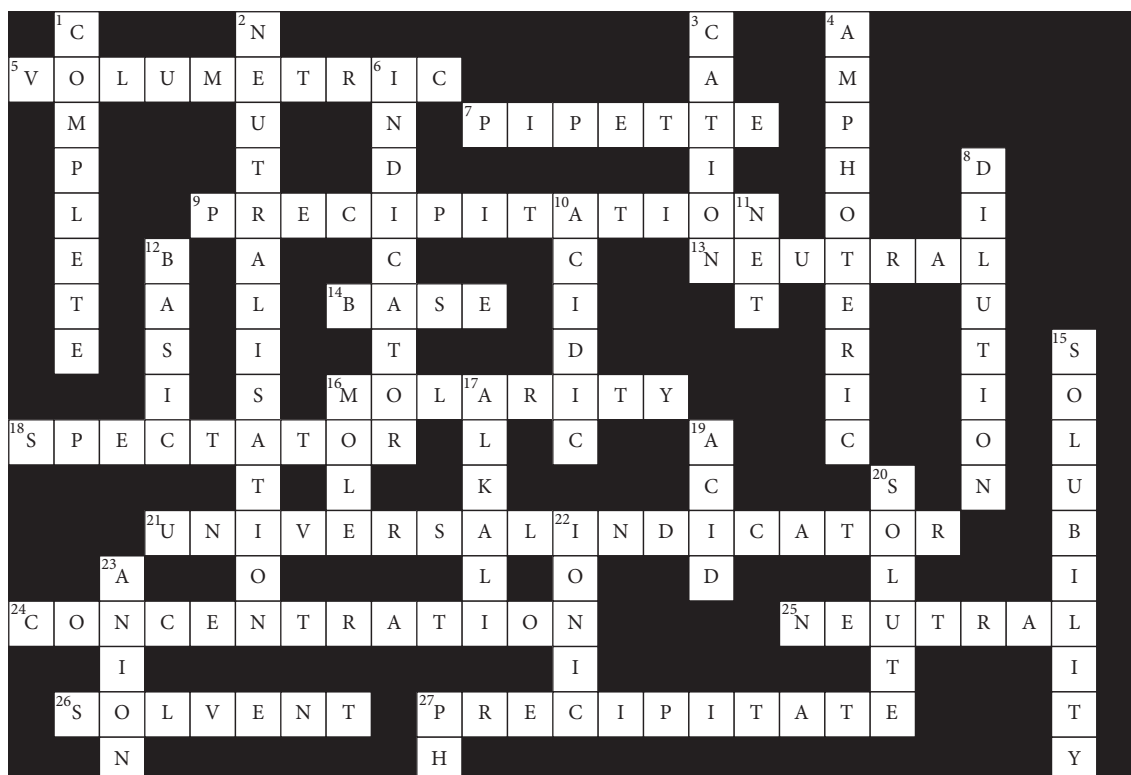


ACTIVITY SHEET ANSWERS**Chapter 7 Suggested answers****Prior knowledge**

- 1 A solute is the solid that dissolves in the liquid solvent.
- 2 Aqueous solutions are solutions of substances dissolved in water.
- 3 The ions separate from each other and become surrounded by water molecules.
- 4 The amount of substance that will dissolve in a given volume of water at a specified temperature.
- 5 Cations are positively charged ions and anions are negatively charged ions.
- 6 Some combinations of dissolved ions react to produce a solid, called a precipitate.
- 7 A complete ionic equation shows all ions present in a reaction, but a net ionic equation does not include spectator ions.
- 8 You can use it to predict whether a precipitate will form when different solutions are mixed.
- 9 As the amount of solute present in a specified amount of solution.
- 10 A mole is Avogadro's number of particles (ions or molecules).
- 11 Molarity is the number of moles of solute per litre of solution.
- 12 A concentrated solution has many dissolved particles, a dilute solution has few.
- 13 Acids have a sour taste and sting cuts. They conduct electricity in solution and turn blue litmus red.
- 14 Bases have a bitter taste and a slippery feel. They conduct electricity in solution and turn red litmus blue.
- 15 Metal oxides and hydroxides
- 16 Acids produce hydrogen ions; bases remove hydrogen ions.
- 17 Acidic and basic solutions
- 18 A soluble base
- 19 An acid reacts with a base to give a salt and water.
- 20 pH measures how acidic or basic a solution is. It is based on the concentration of hydrogen ions.
- 21 The lower the pH the more acidic a solution; the higher the pH the more basic a solution.
- 22 pH paper is quick to use but not as accurate as a pH meter, which needs maintenance.
- 23 Acid + metal → salt + water
- 24
$$\frac{\text{Acid} + \text{carbonate}}{\text{hydrogen carbonate}} \rightarrow \text{salt} + \text{water} + \text{carbon dioxide}$$

Activity sheet 7.1



Activity sheet 7.2

Cations Anions										
	Br ⁻	Cl ⁻	CO ₃ ²⁻	I ⁻	NO ₃ ⁻	O ²⁻	OH ⁻	S ²⁻	SO ₄ ²⁻	PO ₄ ³⁻
Ag ⁺	P	P	P	P	NP	P	P	P	P	P
Al ³⁺	NP	NP	P	NP	NP	P	P	P	NP	P
Ba ²⁺	NP	NP	P	NP	NP	NP	NP	NP	P	P
Ca ²⁺	NP	NP	P	NP	NP	NP	NP	NP	P	P
Cu ²⁺	NP	NP	P	NP	NP	P	P	P	NP	P
Fe ²⁺	NP	NP	P	NP	NP	P	P	P	NP	P
Fe ³⁺	NP	NP	P	NP	NP	P	P	P	NP	P
Mg ²⁺	NP	NP	P	NP	NP	P	P	NP	NP	P
Na ⁺	NP	NP	NP	NP	NP	NP	NP	NP	NP	NP
NH ₄ ⁺	NP	NP	NPH	NP	NP	NP	NP	NP	NP	NP
Pb ²⁺	P	P	P	P	NP	P	P	P	P	P
Sr ²⁺	NP	NP	P	NP	NP	NP	NP	NP	P	P
Zn ²⁺	NP	NP	P	NP	NP	P	P	P	NP	P

Activity sheet 7.4

Part A: Preparing a solution

- 1 Weigh 2.5 g of potassium chloride in a beaker using an electronic balance, then transfer the solute to a 100 mL volumetric flask.
- 2 Rinse the beaker that contained the potassium chloride with distilled water and add the rinse water to the volumetric flask.
- 3 Add more distilled water to dissolve the solute, then fill the flask up to calibration mark.
- 4 Insert the stopper in the flask and invert it several times to ensure all of the solute is dissolved and thoroughly mixed.

Part B: Diluting a solution

- 1 Use a 25 mL pipette to measure that volume of the 1 M solution of copper sulfate.
- 2 Transfer this volume to a 250 mL volumetric flask.
- 3 Add sufficient distilled water to the volumetric flask to make up the solution to the calibration mark.
- 4 Insert the stopper in the flask and invert it several times to ensure the solution is thoroughly mixed.

Chapter 7 Revision

- 1 a $\rightarrow V$, b $\rightarrow Q$, c $\rightarrow O$, d $\rightarrow H$, e $\rightarrow P$, f $\rightarrow T$, g $\rightarrow R$, h $\rightarrow K$, i $\rightarrow D$, j $\rightarrow A$, k $\rightarrow J$, l $\rightarrow B$, m $\rightarrow E$, n $\rightarrow I$, o $\rightarrow S$, p $\rightarrow L$, q $\rightarrow M$, r $\rightarrow N$, s $\rightarrow C$, t $\rightarrow U$, u $\rightarrow G$, v $\rightarrow F$
- 2 a An *acidic solution* has a pH of less than 7. A *basic solution* has a pH greater than 7
 b A *neutral species equation* shows all reactants and products as electrically neutral compounds. A *complete ionic equation* shows all ions present in the solution including spectator ions. A *net ionic equation* shows only the reacting ions and any compound produced or consumed.
 c *Acids* are substances that produce hydrogen ions in aqueous solution; *bases* are substances that contain oxide or hydroxide ions, or produce hydroxide ions in aqueous solution. *Alkalis* are soluble bases.
- 3 a Magnesium carbonate, MgCO_3
 b The new combination of magnesium and carbonate ions would result in a precipitate as magnesium carbonate is insoluble. (The other new combination of sodium and chloride ions would not result in a precipitate as sodium chloride is soluble.)
 c $\text{Na}_2\text{CO}_3(\text{aq}) + \text{MgCl}_2(\text{aq}) \rightarrow \text{MgCO}_3(\text{s}) + 2\text{NaCl}(\text{aq})$
 d $2\text{Na}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) + \text{Mg}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow \text{MgCO}_3(\text{s}) + 2\text{Na}^+(\text{aq}) + 2\text{Cl}^-(\text{aq})$
 e $\text{CO}_3^{2-}(\text{aq}) + \text{Mg}^{2+}(\text{aq}) \rightarrow \text{MgCO}_3(\text{s})$
- 4 a Lead(II) chloride, PbCl_2
 b The new combination of lead(II) and chloride ions would result in a precipitate as lead(II) chloride is insoluble. (The other new combination of aluminium and nitrate ions would not result in a precipitate as all nitrates are soluble.)
 c $2\text{AlCl}_3(\text{aq}) + 3\text{Pb}(\text{NO}_3)_2(\text{aq}) \rightarrow 3\text{PbCl}_2(\text{s}) + 2\text{Al}(\text{NO}_3)_3(\text{aq})$
 d $2\text{Al}^{3+}(\text{aq}) + 6\text{Cl}^-(\text{aq}) + 3\text{Pb}^{2+}(\text{aq}) + 6\text{NO}_3^-(\text{aq}) \rightarrow 3\text{PbCl}_2(\text{s}) + 2\text{Al}^{3+}(\text{aq}) + 6\text{NO}_3^-(\text{aq})$
 e $\text{Pb}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow \text{PbCl}_2(\text{s})$

5 a Silver nitrate and sodium chloride solutions

- b A precipitate of solid silver chloride forms.
- c The silver and chloride ions are more strongly attracted to each other than they are to the water molecules that surround them, so they cluster together in groups to form a solid, which then precipitates out of the solution.
- d The sodium and nitrate ions are less strongly attracted to each other than they are to the water molecules that surround them, so they do not cluster together to form a solid. And all nitrates are soluble.
- e $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$

6 a Write the relevant relationship, $c = \frac{n}{V}$.

- b Identify the known quantities (n and V) and check the units (should be mol and L).
- c Substitute into the formula to find concentration, c .
- d Complete the calculation and round the answer to the appropriate number of significant figures.

7 So that the correct mole ratio of species in the reaction is known.

8 $c_1 V_1 = c_2 V_2$ so $c_2 = \frac{c_1 V_1}{V_2} = 0.5 \times \frac{50}{250} = 0.1 \text{ mol L}^{-1}$

9 In aqueous solution, acids produce hydrogen ions and bases contain or produce hydroxide ions. The hydroxide ions react with hydrogen ions to form water. As bases therefore remove hydrogen ions from a solution they are the chemical opposite of acids.

10 a Hydrogen power

- b The concentration of hydrogen ions decreases by a factor of 10 each step up the pH scale. The concentration of hydrogen ions is equal to 10 to the power of the pH value but with a negative sign inserted.
- c The concentration of hydroxide ions increases by a factor of 10 each step up the pH scale. The concentration of hydroxide ions is equal to 10 to the power of 14 minus the pH value.
- d As the concentration of one ion rises the other falls. The exponents (superscripts) of the two concentrations always add up to -14 .

11 a $2\text{HNO}_3(\text{aq}) + \text{Ca}(\text{s}) \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{H}_2(\text{g})$

b $2\text{HCl}(\text{aq}) + \text{Na}_2\text{CO}_3(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$

c $\text{H}_2\text{SO}_4(\text{aq}) + 2\text{LiHCO}_3(\text{aq}) \rightarrow \text{Li}_2\text{SO}_4(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) + 2\text{CO}_2(\text{g})$